Gradient-Boosted Trees and Consensus-Risk

For Opioid and Polypharmacy Emergency Department Visits in Virginia's APCD

R. Jerome Dixon, MDA, MSc IT

Integrative Life Sciences Doctoral Program (Home Department)

School of Pharmacy (Department of Pharmacotherapy & Outcomes Science · Research Lab)

Virginia Commonwealth University

November 9, 2026

VCU School of Life Sciences and Sustainability | VCU School of Pharmacy

Supervisory Committee

INTEGRATIVE LIFE SCIENCES DOCTORAL PROGRAM | SCHOOL OF PHARMACY | DEPARTMENT OF PHARMACOTHERAPY & OUTCOMES SCIENCE

COMMITTEE CHAIR & DISSERTATION SPONSOR

Elvin T. Price, Pharm.D., Ph.D., FAHA
Associate Professor, Department of Pharmacotherapy & Outcomes Science · VCU School of Pharmacy

SUPERVISORY COMMITTEE MEMBERS

Franklin Bost, BPD, MBA, IDSA, FAIMBE
Professor Emeritus, Dept. of Biomedical Engineering & College of Engineering

Tamas Gal, Ph.D.
Director of Research Informatics & Core Director, Massey Comprehensive Cancer Center

Dayanjan S. Wijesinghe, Ph.D.
Associate Professor, VCU School of Pharmacy

Jonathan P. DeShazo, Ph.D.
Director of Real World Evidence Generation, CVS Health

Roadmap

QUALIFYING EXAM AGENDA | 5 CORE MODULES

I. Problem

II. Governance & Data

III. Cohorts

IV. Modeling

V. Translation & Deployment

Downstream Clinical & Economic Effects

MODULE I | INFOGRAPHIC PART 1 OF 4 · DOWNSTREAM PUBLIC HEALTH & ECONOMIC TOLL

>500,000 US Opioid Deaths & $78.5B Annual Cost



Since 1999, prescription opioid overdoses and abuse have claimed over half a million lives in the US, generating an economic burden exceeding $78.5 billion annually in healthcare, addiction treatment, and criminal justice expenditures.

Geriatric Polypharmacy: 125,000 Deaths & 700,000 ED Visits Annually

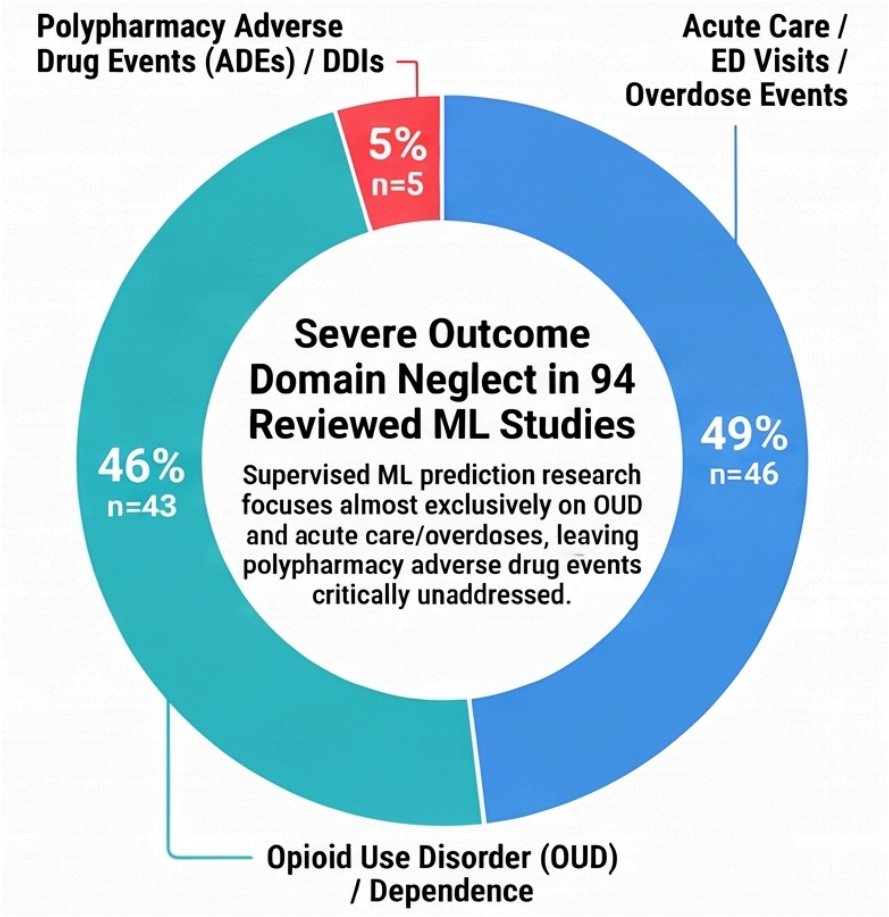


Concurrent use of 5 or more medications affects over 40% of adults aged 65 and older, making polypharmacy the leading driver of preventable adverse drug events in older adults.

Figure 1A (Part 1): Downstream clinical and economic effects of US opioid overdoses and geriatric polypharmacy (CDC WONDER; Florence et al., *Med Care*; Charlesworth et al., *J Gerontol A*; Budnitz/Shehab et al., *JAMA*).

Opioid Research Overshadows Polypharmacy

MODULE I | INFOGRAPHIC PART 2 OF 4 · SYSTEMATIC REVIEW (N = 94 ML STUDIES)





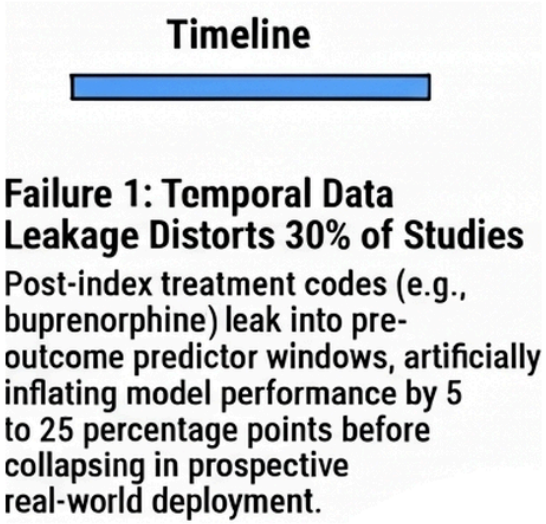
Claims-Based Opioid ED Modeling Void

Out of 1,473 published OUD literature records, only 6 studies specifically evaluate machine learning models designed to predict opioid-related emergency department visits using administrative claims data.

Figure 1A (Part 2): Published outcome domain distribution across 94 supervised ML studies and the claims-based opioid ED modeling void.

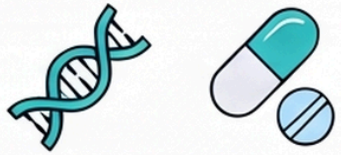
ML Gap – No PGx features. No XAI. No Target Leakage Validation.

MODULE I | INFOGRAPHIC PART 3 OF 4 · PROBAST APPRAISAL (N = 73 MODELS)



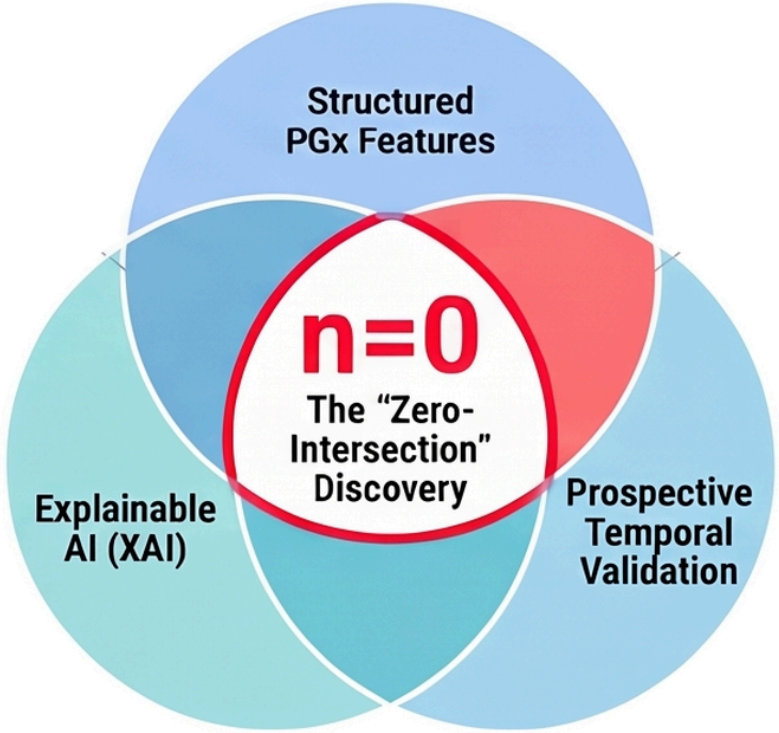
Failure 1: Temporal Data Leakage Distorts 30% of Studies

Post-index treatment codes (e.g., buprenorphine) leak into pre-outcome predictor windows, artificially inflating model performance by 5 to 25 percentage points before collapsing in prospective real-world deployment.



Failure 2: 81% Omit Pharmacogenomics (PGx)

Despite mature CPIC guidelines and known gene-drug modifiers (e.g., CYP2D6, OPRM1), 81% of reviewed ML prediction pipelines completely exclude structured pharmacogenomic features.



Across 73 PROBAST-assessed modeling studies, zero published systems successfully combine structured PGx features, explainable AI (XAI), and prospective temporal validation.

Figure 1A (Part 3): PROBAST appraisal of 73 predictive models exposing the ML gap ($n = 0$ combining PGx features, XAI, and target leakage / temporal holdout validation).

90% of ADEs Occur Within 21 Days

MODULE I | STEP 1

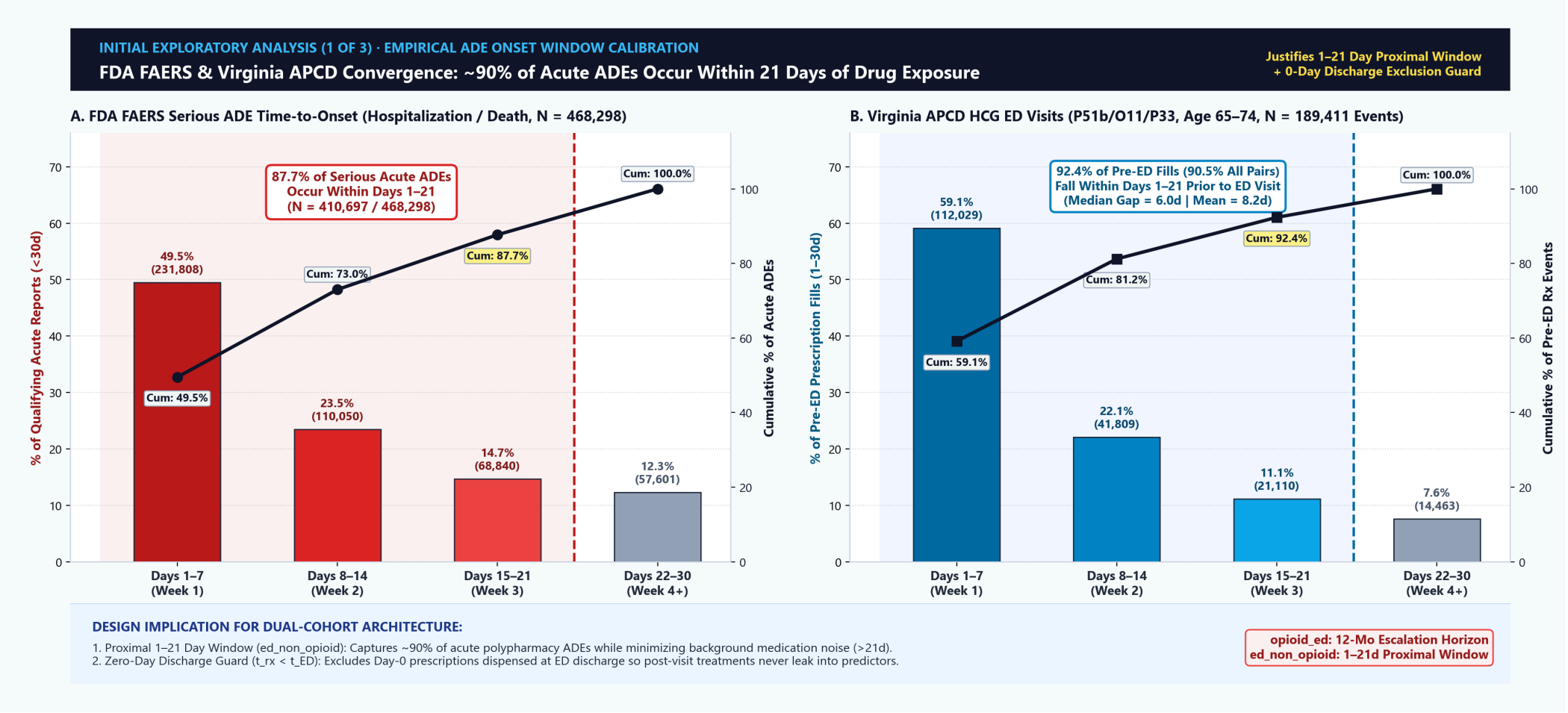


Figure 1B (Part 1): Convergent time-to-onset distribution across FDA FAERS ($N = 468,298$, 87.7% $\leq$ 21d) and Virginia APCD HCG ED visits ($N = 189,411$, 92.4% $\leq$ 21d).

Initial Feature Analysis

MODULE I | STEP 3A

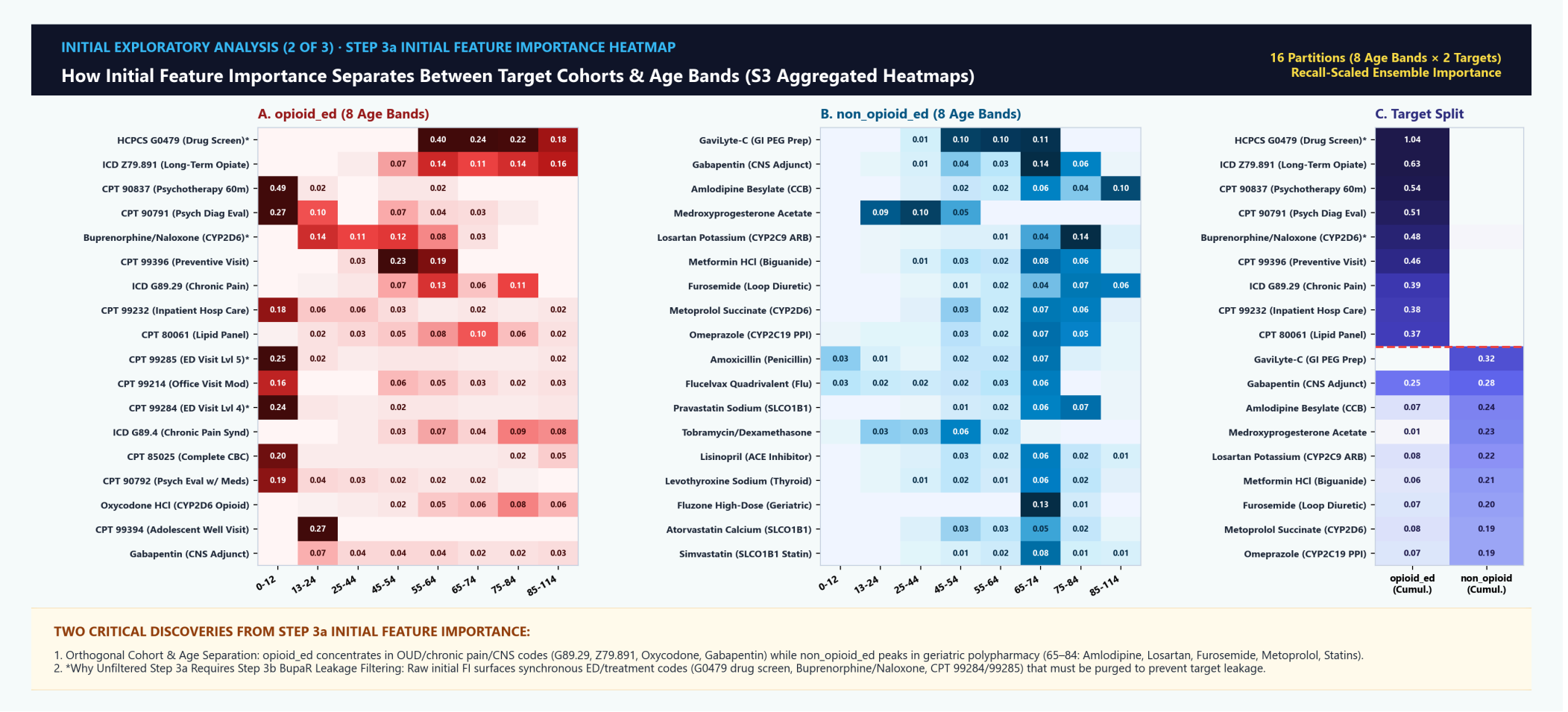


Figure 1B (Part 2): Step 3a multi-model (`CatBoost` , `XGBoost` , `XGBoost-RF`) aggregated feature importance across all 8 age bands (`0-12` to `85-114`) and both targets (`opioid_ed` vs. `non_opioid_ed`).

CPIC Drugs Drive Initial Feature Importance

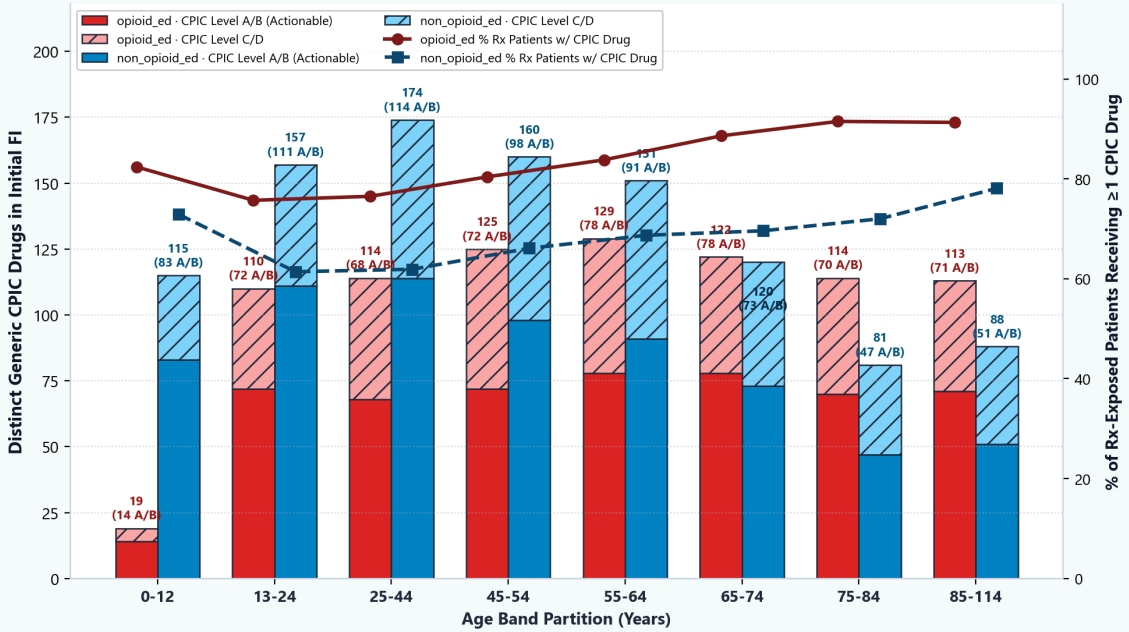
MODULE I | STEP 5

INITIAL EXPLORATORY ANALYSIS (3 OF 3) · CPIC PHARMACOGENOMIC DRUGS IN INITIAL FEATURE IMPORTANCE

Number of CPIC-Actionable Drugs Surfaced in Initial Feature Importance by Target Cohort & Age Band

opioid_ed: 187 Distinct CPIC Drugs (333 Features)
non_opioid_ed: 224 Distinct CPIC Drugs (439 Features)

A. Distinct CPIC Drugs in Initial Feature Importance by Target Cohort & Age Band



B. Dominant CPIC Gene–Drug Pairs in Initial Feature Importance by Target

Pharmacogene	CPIC Lvl	opioid_ed Top FI Drugs (187 CPIC Drugs | 76–92% Rx Pts)	non_opioid_ed Top FI Drugs (224 CPIC Drugs | 61–78% Rx Pts)
CYP2D6 (Analgesics, SSRIs, β-Blockers)	Level A / B	• Oxycodone (#17), Hydrocodone (#14) • Tramadol (#39), Codeine, Ondansetron • Venlafaxine, Risperidone, Aripiprazole	• Tramadol (#3), Hydrocodone (#6) • Metoprolol (#8), Carvedilol, Duloxetine • Ondansetron, Amitriptyline, Mirtazapine
CYP2C19 (PPIs, SSRIs, Antiplatelets)	Level A	• Omeprazole (8/8 bands), Pantoprazole • Sertraline, Citalopram, Escitalopram • Diazepam (8/8 bands), Amitriptyline	• Omeprazole (#9), Pantoprazole (#12) • Escitalopram (#21), Sertraline • Citalopram, Clopidogrel
CYP2C9 & VKORC1 (NSAIDs, Sulfonyl-ureas, Warfarin)	Level A / B	• Ibuprofen (8/8 bands), Meloxicam • Celecoxib, Naproxen, Diclofenac • Phenytoin (CYP2C9 / HLA-B)	• Losartan (#5), Diclofenac, Ibuprofen • Meloxicam, Celecoxib, Glimepiride • Warfarin (CYP2C9 / VKORC1)
SLCO1B1 & ABCG2 (Statins / Hepatic Transporters)	Level A	• Atorvastatin, Simvastatin • Pravastatin, Rosuvastatin • Co-prescribed in ages 45–85+	• Pravastatin (#12), Atorvastatin (#17) • Simvastatin (#18), Rosuvastatin • High geriatric polypharmacy importance
OPRM1 / COMT / UGT2B15 / G6PD	Level B / C	• Buprenorphine/Naloxone (#5) • Fentanyl (#24), Lorazepam (8/8 bands) • Sulfamethoxazole/TMP (8/8 bands)	• Lorazepam (#14), Ofloxacin (#5) • Ciprofloxacin, Sulfamethoxazole/TMP • Nitrofurantoin (G6PD)

EMPIRICAL BRIDGE TO STEP 5 PGx FEATURE ENGINEERING (pgx_num_drugs & pgx_num_cpic_drugs):

1. High CPIC Density in Top Predictors: 24%–35% of all drug features in initial Feature Importance are CPIC-actionable medications (187 in opioid_ed; 224 in non_opioid_ed).
2. Calibrated Scientific Framing: Claims capture CPIC-actionable medication exposure burden (identifying patients whose regimen warrants PGx review), NOT patient genotypes.

pgx_num_drugs & pgx_num_cpic_drugs
Rank #1–#3 Across Final Models

Figure 1B (Part 3): Number of distinct CPIC-actionable drugs in initial Feature Importance results by target cohort (`opioid_ed` vs. `non_opioid_ed`) and age band.

The Frontline Clinical Bottleneck

MODULE I | INFOGRAPHIC PART 4 OF 4 · POINT-OF-CARE TRANSLATION GAP

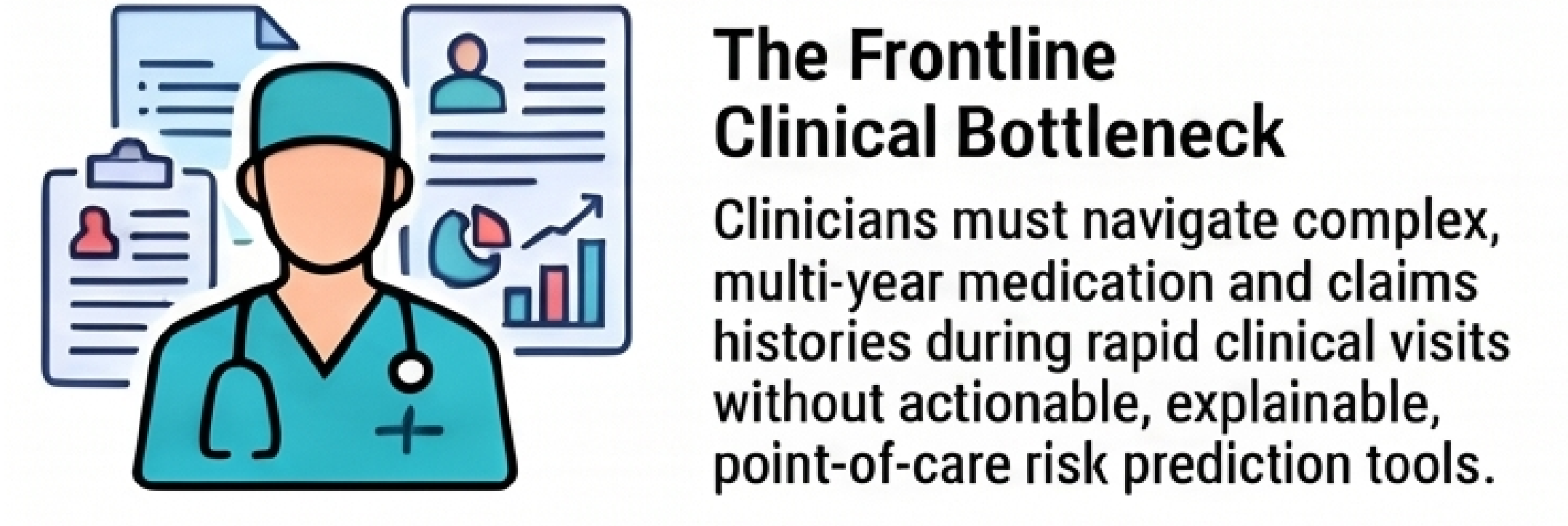


Figure 1A (Part 4): The Frontline & Predictive Clinical Bottleneck across longitudinal medication and administrative claims histories.

AIMS

MODULE I | AIM 1 · COHORT 1 (`opioid_ed`) | AIM 2 · COHORT 2 (`ed_non_opioid`) | AIM 3 · DECISION SUPPORT

Aim 1 · Cohort 1 Opioid ED Prediction & Attribution (Ch. 3)

Develop and temporally validate `CatBoost`, `XGBoost`, and `XGBoost-RF` models with $\mathrm{SHAP} \cap \mathrm{FFA}$ consensus attribution for **Cohort 1** (`opioid_ed` , 12–mo lookback, ages 13–64), evaluating pre-index leakage prevention, discrimination (`PR-AUC` , `AUROC`), and calibration on an independent **2019 temporal holdout**.

Aim 2 · Cohort 2 Polypharmacy & DDI Attribution (Ch. 4)

Develop and temporally validate `CatBoost`, `XGBoost`, and `XGBoost-RF` models with $\mathrm{SHAP} \cap \mathrm{FFA}$ Boolean Horn rules for **Cohort 2** (`ed_non_opioid` , 21–day lookback, ages 65–114) to identify reproducible risk predictors, candidate multi-drug interactions (`IE`), and consensus risk attenuation priorities (`IR`) on the **2019 holdout**.

Aim 3 · Usable Decision-Support Tool (Ch. 5)

Translate validated Aim 1 and Aim 2 risk estimates and interpretable Boolean rules into a usable prototype decision-support tool for pharmacist-led Medication Therapy Management (MTM) review prioritization—evaluating explanation fidelity, clinical interpretability, and workflow usability rather than claiming clinical effectiveness.

Cohort 1: Opioid-Related ED Events (`opioid_ed` , Ages 13–64)

- **Pathophysiology:** Progressive μ-opioid receptor tolerance, escalating MME titration, and synergistic CNS depressants (benzodiazepines, gabapentinoids) blunting respiratory drive.
- **Lookback Horizon: 12–Month Pre–Index Window** capturing chronic prescription escalation and behavioral comorbidities.
- **Claims Anchor (t_0):** Direct diagnostic billing for `F11.20` (opioid dependence) and `T40.0X` – `T40.4X` (opioid poisonings) scanned across **all 10 claim diagnostic positions**.

Cohort 2: Geriatric Polypharmacy & "Silent ADEs" (`ed_non_opioid` , Ages 65–114)

- **Why Pairwise CDS Fails:** Pairwise DDI ($A + B$) checkers trigger **90%–95% alert fatigue** while missing higher-order ($A + B + C + D$) synergies and age-related renal/hepatic decline.
- **The "Silent ADE" Coding Reality:** Fewer than **5%** of drug-induced ED visits are billed with explicit poisoning codes (`T36` – `T50`). Hospitals bill the **downstream clinical syndrome**:
  - **Falls & Syncope:** `R29.6` (falls), `S72` (hip fracture), `R55` (syncope) ← *Z-drugs, benzos, α-blockers*.
  - **Delirium, Bleeds & AKI:** `F05` (delirium), `K92.2` (GI bleed), `N17` (Triple-Whammy AKI).
- **Claims Anchor (t_0):** Milliman HCG emergency lines (`P51b` , `O11` , `P33`) anchored to a **21–Day Proximal Rx Window** (`0-day` guard), with **0 opioid codes** in study year.

Methodological Demarcation: Modeling Cohort 1 (12–month chronic opioid trajectory) and Cohort 2 (21–day proximal geriatric polypharmacy) separately prevents conflating distinct pathophysiologies, age distributions, and coding mechanics.

Governance and Frameworks

MODULE II | PHASE 1 · CRISP-DM, TRIPOD+AI & CLINICAL OODA LOOP

1. CRISP-DM (Systems Lifecycle)

- 6 iterative engineering phases governing the 10-step APCD pipeline.
- Ingestion → Cohort → Whitelist → Model → Eval → Microservice.

2. TRIPOD+AI (Biomedical Validation)

- Transparent reporting; Type 2b / Type 3 temporal validation.
- 2016–2018 train, untouched 2019 calendar holdout (zero post-hoc recalibration).

3. Clinical OODA Loop (Translation)

- **Observe / Orient:** GBT models & formal Boolean Horn rules.
- **Decide / Act:** Decision Support Tool

Appendix A6: Master Acronym & Framework Glossary (CRISP-DM, TRIPOD+AI, OODA) →

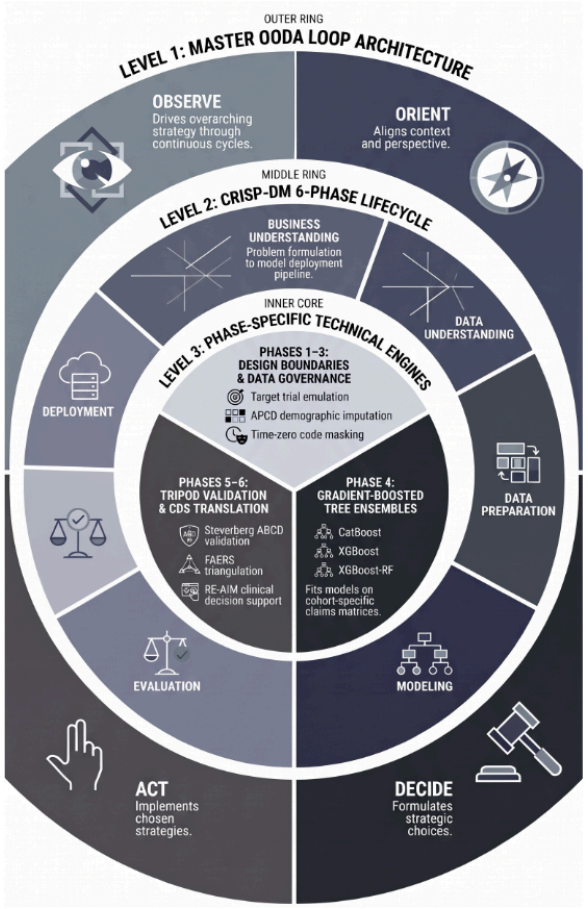


Figure 2: Three-Tier Governance Architecture: Master OODA Loop enclosing 6-phase CRISP-DM and phase-specific technical validation engines.

Why Virginia APCD Over EHR

MODULE II | PHASE 1 · STATEWIDE GENERALIZABILITY & DATA ARCHITECTURE

Virginia APCD (Selected)	**Single-System EHR** (Rejected)
Statewide Capture: Follows patients across health systems (eliminating out-of-network ED blindness).	**Fragmentation:** High loss-to-follow-up when crossing systems. Misclassifies out-of-network overdoses as "healthy non-events."
Medication Reality: Captures adjudicated pharmacy fills, proving physical drug possession.	**Intent vs Exposure:** Captures written prescriptions, missing ~30% primary non-adherence drop-offs.
Linkage: Single honest broker (VHI) guarantees zero probabilistic linkage attrition.	

The Methodological Compensation

We acknowledge the lack of molecular genotypes and real-time vitals in claims by computing CPIC PGx-burden proxies (*CYP2D6/CYP2C19* substrate counts) directly from longitudinal co-prescriptions.

Figure 3: Data Source Architecture: Statewide capture and adjudicated dispenses in Virginia APCD overcome single-system EHR fragmentation and prescription abandonment.

Why Virginia APCD Is Selected

- **Statewide Capture:** Follows patients across health systems, eliminating out-of-network ED blindness.
- **Medication Reality:** Captures paid/adjudicated dispenses, proving physical drug possession.
- **Linkage Integrity:** Single trusted data steward (VHI) guarantees 0 probabilistic linkage attrition.

The Methodological Compensation

- **EHR Genotype Gap:** APCD lacks molecular genotypes and real-time bedside vitals.
- **CPIC Guideline Mapping & Drug Counts:** We compensate by mapping dispensed APCD drugs to whether they carry a CPIC guideline (`drug_cpic_mapping_global.csv`) and counting patient-level total vs. CPIC drugs (`pgx_num_drugs` , `pgx_num_cpic_drugs`) and how many patients carry CPIC drugs.

Medical & Pharmacy Claims Schema

MODULE III | STEP 2

Medical & Pharmacy Relational Join (`mi_person_key`)

- **Medical (`25` cols):** 10 ICD-10 Dx (`primary..ten_icd_diagnosis_code`), 9 ICD-Proc + 3 CPT (`procedure_code` , `cpt_mod_1/2`), HCG (`hcg_line` , `hcg_detail`), imputed demographics.
- **Pharmacy:** Paid dispenses (`incurred_date → event_date`), generic RxNorm `drug_name` , `therapeutic_class_1` .
- **Unified Timeline (`unified_event_fact_table`):** Linked on `(mi_person_key, event_year, age_band)` with demographic backfill & global `event_sequence` ordering.

Key Pre-Index Feature Families ($t < t_0$)

- **Clinical & Rx Indicators:** `item_icd_*` (10 Dx positions), `item_cpt_*` , `item_drug_*` (active generic fills & concurrency).
- **CPIC Drug Mapping & Utilization (`5_pgx_analysis`):** Maps APCD drugs to CPIC guideline status to count patient drugs (`pgx_num_drugs` , `pgx_num_cpic_drugs`) & patients with CPIC drugs, plus `n_events` density strata (`P25/P50/P95`).

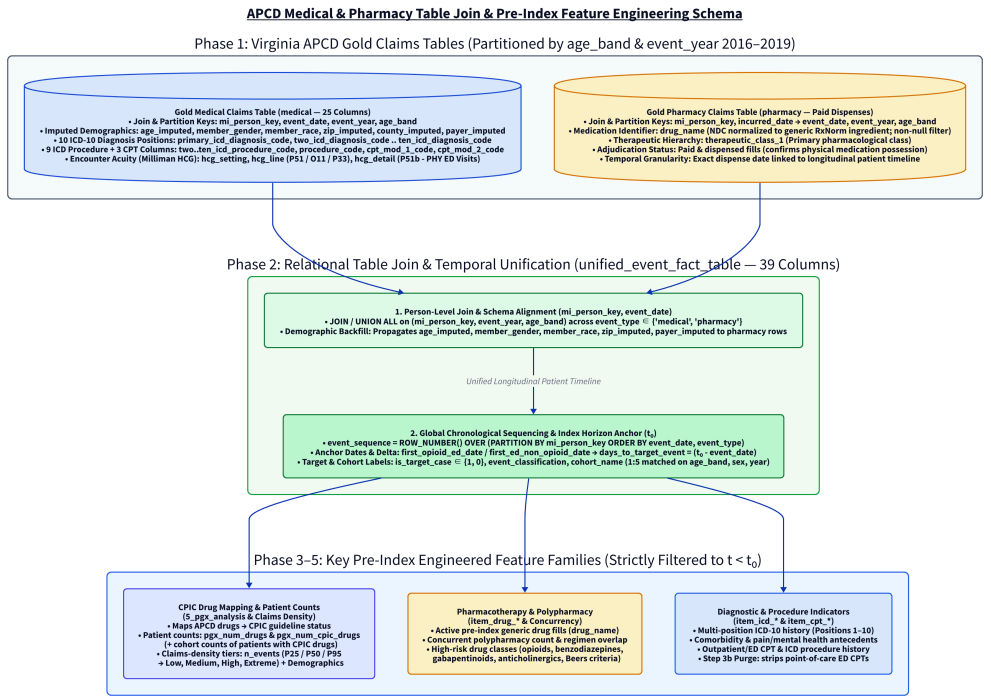


Figure 4: Relational unification of APCD Gold `medical` and `pharmacy` claims into the 39-column `unified_event_fact_table` and pre-index feature matrix.

Schema Integration: Deterministic linkage on `mi_person_key` and global chronological sequencing (`event_sequence`) align diagnoses, procedures, and paid drug fills prior to t_0 .

Cohort 1 Pipeline: Opioid-Related ED (`opioid_ed`)

MODULE III | STEP 2

Clinical Anchor (t_0)

- **ICD-10** `F11.20` (Opioid dependence).
- **Acute Poisonings:** `T40.0X` – `T40.4X`.
- Scanned across all 10 claim positions.

Step 3b Leakage Purge

- Partitions candidate events around t_0 ($t < t_0$ vs. $t \geq t_0$).
- Purges **post-index rescue & MAT induction** (buprenorphine, naloxone) & toxicology screens.
- Locks symmetric pre-index antecedent whitelist.

1:5 Matching & DuckDB QA

- Controls matched on 8 age bands, sex, year.
- Asserts single index date per patient-year.

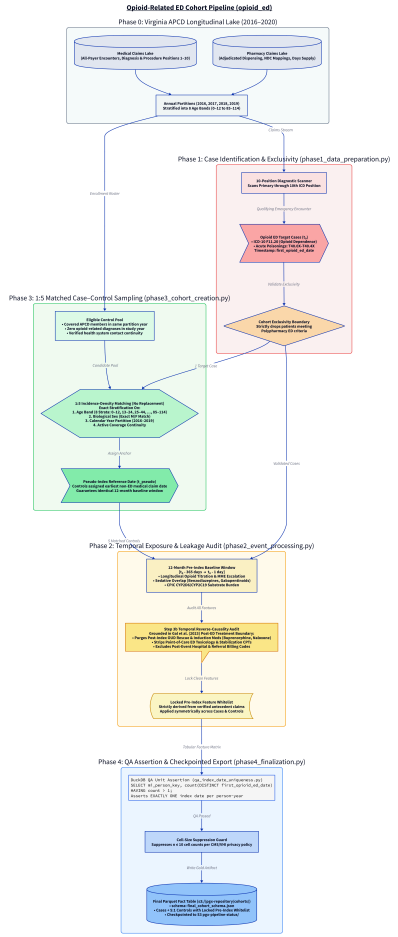


Figure 4A: Opioid-Related ED Cohort Derivation Pipeline (APCD 2016–2020) with 12-month pre-index lookback and locked antecedent whitelist.

Cohort 2 Pipeline: Polypharmacy ADEs (`ed_non_opioid`)

MODULE III | STEP 2

Clinical Anchor (t_0)

- **Milliman HCG Codes:** `P51b`, `O11`, `P33`.
- **Absolute Opioid Cleanse:** Zero opioid ICD codes across all 10 positions in study year.

Sequential ADE Funnel

1. **Utilization Cap:** < 7 ED visits/yr.
2. **21-Day Proximity:** Active Rx 1–21d pre-ED (FDA FAERS 90.5%).
3. **0-Day Discharge Guard:** Purges Day 0 fills.

Feature Representation

- 21–30 day acute concurrent exposure.
- Beers criteria & multi-drug regimens.

Cohort Validation FIGURE 1B ↗

- **FAERS vs APCD ADE Events:** >90% kinetic onset convergence at 21 days.
- *Click card to pop out Figure 1B.*

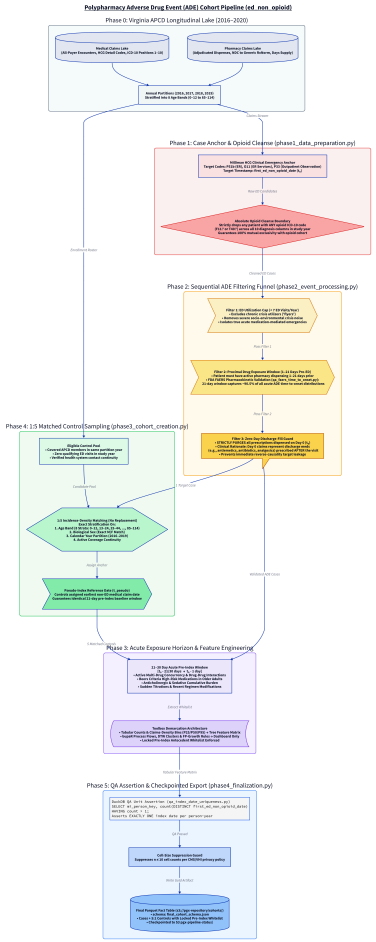


Figure 4B: Polypharmacy Adverse Drug Event Cohort Derivation Pipeline with sequential ADE filtering funnel and 0-day discharge guard.

Gradient-Boosted Trees & Model Selection

MODULE IV STEP 6

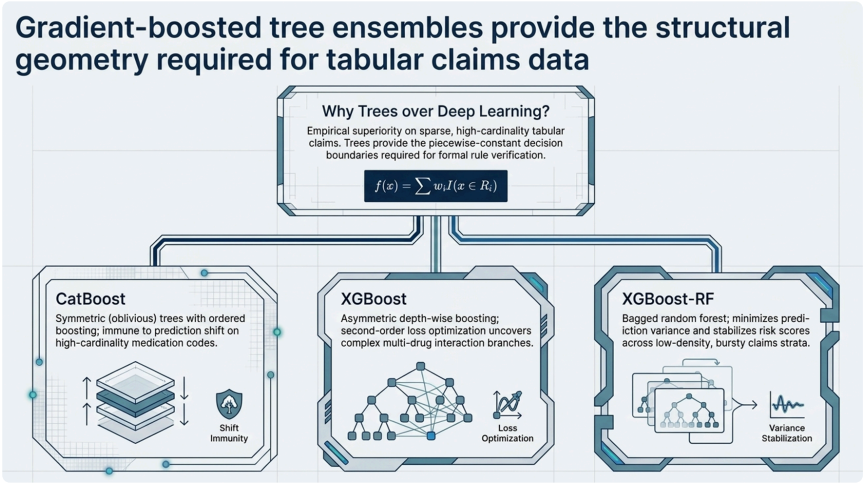


Figure 5: Tri-Model Structural Geometry: Piecewise-constant decision boundaries $f(x) = \sum w_i \mathbb{I}(x \in R_i)$ enabling cross-architecture SHAP consensus and exact CNF rule verification.

CatBoost (Symmetric)

- Oblivious decision trees with ordered boosting.
- Shift immunity on high-cardinality NDC and ICD-10 diagnostic codes.

XGBoost (Asymmetric)

- Depth-wise gradient-boosted decision trees.
- Second-order Hessian loss uncovers complex multi-drug interaction branches.

XGBoost-RF (Bagged)

- Bagged random forest mode with parallel trees.
- Minimizes prediction variance and stabilizes risk across low-density strata.

Step 6 Selection Protocol: Within each cohort, age band, and 4-tier density bin, 25-split MC-CV ranks CatBoost, XGBoost, XGBoost-RF, and Ensemble by **mean PR-AUC** (primary) and **Recall at 0.5** (tiebreaker) to determine the top-performing model for risk scoring and select the winning XGBoost variant plus CatBoost for Step 7 consensus attribution.

Consensus Risk Attribution

MODULE IV | CONSENSUS RISK ARCHITECTURE

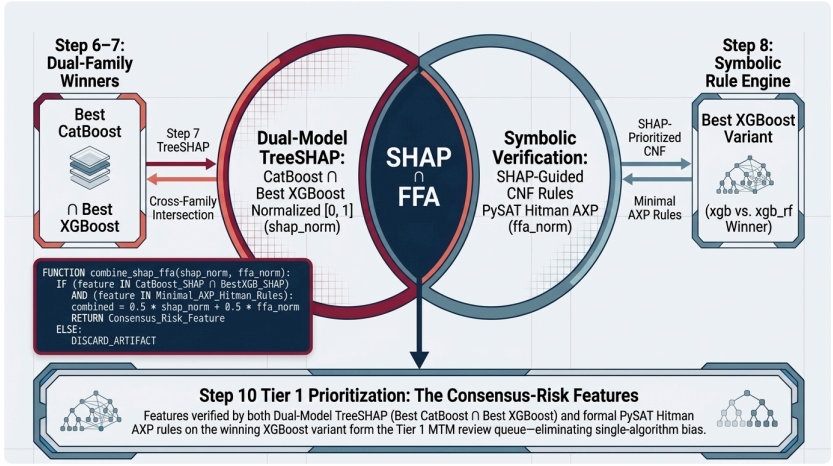


Figure 6A: Aim 2 Consensus-Risk Architecture: Dual-Model TreeSHAP Intersection (`CatBoost` ∩ `Best XGBoost`) guiding Symbolic CNF Rule Extraction on `best_xgb_variant` and Consensus Synthesis.

Dual SHAP Gate & Symbolic CNF

- **Dual-Model `TreeSHAP` Gate:** Computes exact `TreeSHAP` on **best CatBoost** and **best XGBoost variant** (`xgb` vs. `xgb_rf`), enforcing strict feature intersection (`CatBoost` ∩ `Best XGBoost`) and normalized mean score.
- **CNF Compilation:** Explodes root-to-leaf paths of `best_xgb_variant` into Conjunctive Normal Form (CNF) clauses (`_explode_tree_to_dataframe`).
- View Symbolic CNF & PySAT AXP Proof ↗

SHAP-Guided FFA & Synthesis

- **SHAP-Prioritized Rule Extraction:** Rather than extracting all rules from all trees across all models, filters matched CNF rules from `best_xgb_variant` by instance $|\mathrm{SHAP}|$ attribution and active feature coverage → PySAT `Hitman` minimal Abductive Explanations (AXPs; $\mathrm{support} \geq 0.05, \mathrm{confidence} \geq 0.70$).
- **Combined Synthesis:** Ranks drivers via $0.5 \cdot \mathrm{SHAP}_{\mathrm{norm}} + 0.5 \cdot \mathrm{FFA}_{\mathrm{norm}}$ and top-K intersection (`shap_top` ∩ `ffa_top`).

Target Leakage

MODULE IV STEP 3B

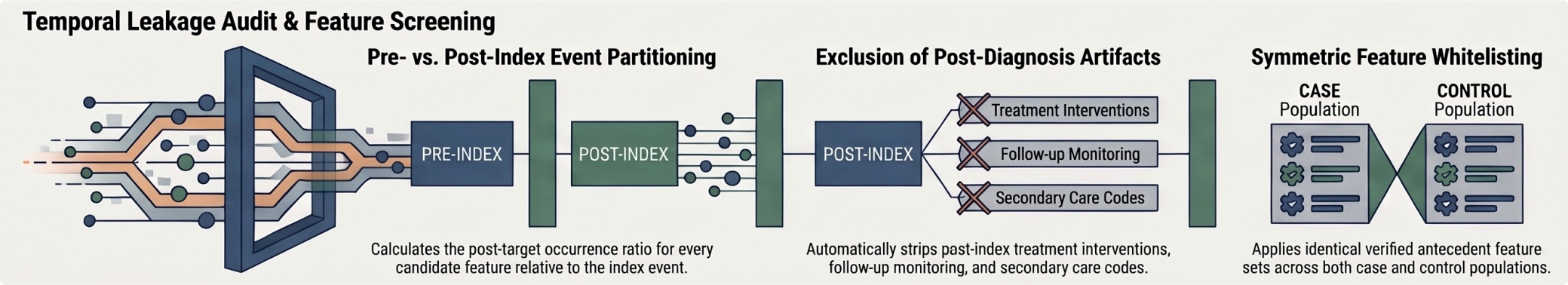


Figure 7: Step 3b Temporal Event-Partitioning Protocol: Pre- vs. post-index ($t < t_0$ vs. $t \geq t_0$) event timing audit, removal of post-event clinical interventions, and symmetric antecedent whitelist locking.

Clinical Rationale (Ochalek, Gal et al., 2023)

- **Post-Overdose Clinical Workflow:** As established in Dr. Gal's VCU ED research (*Ochalek, Gal et al., 2023, DADR*), rescue agents and MAT induction (e.g., buprenorphine, naloxone) are initiated *as a clinical consequence following* an acute overdose.
- **The Reverse-Causality Trap (30% of Literature):** Unmonitored gain rankings (`Step 3a`) capture post-event stabilization and confirmatory toxicology orders occurring at $t \geq t_0$.

Step 3b Symmetric Antecedent Whitelist

- **Temporal Event Partitioning (`bupaR` Audit):** Partitions every candidate feature's timestamps relative to t_0 ($t < t_0$ pre-index vs. $t \geq t_0$ post-index).
- **Automated Post-Target Purge:** Strips candidate codes with $\tau_{\text{post}} \geq 0.80$ (`post_target_ratio ⩾ 0.80` ; post-event MAT induction, ED tox screens, follow-up billing).
- **Immutable Whitelist (`safe_feature_filter.json`):** Locks verified pre-index antecedents symmetrically across cases & controls *(empirical catalogs reserved for Nov 23 Defense)*.
- View Step 3b Event-Partitioning Protocol ↗

Actionable Intelligence and Workflow Integration

MODULE V STEP 10

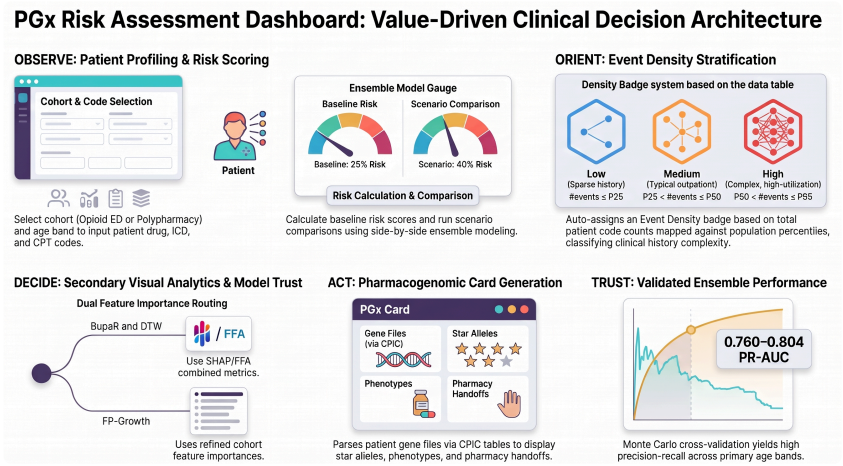


Figure 8: Aim 3 Decision-Support Prototype: Value-Driven Clinical Decision Architecture, OODA Information Delivery, and Event Density Stratification.

Translating Validated Findings for Clinical Review

- **Actionable Triage Prioritization:** Translates Aim 1 risk estimates and Aim 2 consensus features into an interpretable Tier 1–3 medication-review priority list.
- **Explanation Fidelity & Usability:** Evaluates whether concise `IF-THEN` rules accurately reflect model predictions and support rapid pharmacist review.
- **OODA Delivery Framework:** Observe – orient – decide – act framework for delivery of information.
- **Lean Value Stream:** Framework for ranking and layout of information (high to low, left to right by value).

Cognitive Ergonomics vs. 95% Alert Fatigue

- **Overcoming Commercial Pop-ups:** Asynchronous review prioritization replaces **90%–95% overridden** synchronous pop-up interrupts.
- **Sound Symbolic Horn Rule:**

$$\text{IF } [\text{Opioid} \geq 1] \wedge [\text{Benzodiazepine} \geq 1] \wedge [\text{Refill} \leq 14\text{d}] \implies \text{Tier 1 Alert}$$

- **Interactive "What-If" Scenario Exploration:** Displays model-predicted risk shifts when regimen features are toggled while preserving 100% clinician authority.

Appendix A7: Regulatory Demarcation & HIPAA Cloud Architecture →

Dissertation Portfolio & Status

MODULE V | PHASE 6 · DISSERTATION PORTFOLIO & STATUS

Core Empirical Research (Aims 1–3 · Published Papers)

- **Chapter 3 (Specific Aim 1):**
  Temporal Drivers of Opioid ED Encounters in Virginia's APCD
  Status: PUBLISHED · *Clin Transl Sci* (Wiley, 2026; DOI: 10.1111/cts.70690)
  Contribution: Step 3b leakage audit, 12-mo chronic opioid model (`opioid_ed`), 2019 temporal holdout.
- **Chapter 4 (Specific Aim 2):**
  SHAP-Informed Formal Feature Attribution for Polypharmacy ADEs
  Status: PUBLISHED · *Clin Transl Sci* (Wiley, Sep 2026; DOI: 10.1111/cts.70718)
  Contribution: Dual-model `TreeSHAP ∩ FFA` consensus, 21-day acute ADE cohort (`ed_non_opioid`), interaction effects (IE).
- **Chapter 5 (Specific Aim 3):**
  Serverless Clinical Decision Support for Pharmacist Medication Review
  Status: PUBLISHED · *Clin Transl Sci* (Wiley, Oct 2026; DOI: 10.1111/cts.70697)
  Contribution: Zero-PHI cloud prototype on AWS Lambda (<$5/mo), Tier 1–3 triage, overcoming 95% alert fatigue.

Foundational, Methods & Synthesis (Dissertation Status)

- **Chapter 1 (Clinical Foundation & Literature Appraisal):**
  Explainable AI & PGx in High-Risk Prescribing: A Systematic Review
  Status: INVITED REVIEW · *J Pers Med* (MDPI)
  Contribution: Systematic review of 94 ML studies; establishes "The Empty Intersection" (0/73 PROBAST models).
- **Chapter 2 (Scalable Data Architecture & Governance):**
  A Scalable Big-Data Claims Processing Architecture and Governance Lifecycle
  Status: WITH EDITOR FOR REVIEW · *J Big Data* (Springer)
  Contribution: Three-tier governance (CRISP-DM, TRIPOD+AI, Clinical OODA Loop) and DuckDB EC2 claims pipeline (15.1× speedup).
- **Chapter 6 (Translational Synthesis & Dissertation Build):**
  Translational Synthesis, Methodological Defenses, and Clinical Horizons
  Status: COMPLETE DRAFT FOR COMMITTEE REVIEW
  Contribution: Comprehensive synthesis of Chapters 1–5, formal methodological defenses, and prospective clinical validation roadmaps.

Dissertation Status: 3 Published Papers (*Clinical and Translational Science*) • **1 Under Review** (*Journal of Big Data*) • **1 Invited Review** (*Journal of Personalized Medicine*) • **Full Manuscript Drafted** across all 6 chapters for Qualifying Exam and Final Defense.

Study Boundaries

MODULE V | PHASE 6 · 8 INTERLOCKING EPIDEMIOLOGICAL & GOVERNANCE SAFEGUARDS

EPIDEMIOLOGICAL COHORT DESIGN (4 SAFEGUARDS)

1. Absolute Opioid Cleanse

Zero opioid ICD-10 codes across all 10 diagnostic positions in the study year isolates pure geriatric ADEs in `ed_non_opioid`.

2. High-Utilization Cap (<7 ED/yr)

Excludes chronic utilizers with ≥ 7 annual ED visits to prevent high-frequency outlier distortion of feature representations.

3. 21-Day Lookback Window

Active Rx $[t_0 - 21, t_0 - 1]$ empirically validated by convergent FAERS/APCD kinetics (>90% of acute ADEs occur within 21 days).

4. Zero-Day Discharge Guard ($t_{\text{claim}} < t_0$)

Strict timing guard purges day-zero fills (`qa_index_date_uniqueness.py`) and $\tau_{\text{post}} \geq 0.80$ confounders (348 purged).

METHODOLOGICAL & CLINICAL GOVERNANCE (4 SAFEGUARDS)

5. Aim 3: Decision Support, Not an Outcomes Trial

Translates validated model findings into an interpretable prototype tool evaluating explanation fidelity; prospective clinical trials are out of scope. App. A7 ↗

6. Consensus Risk Attribution, Not Causation

Observational claims and `SHAP ∩ FFA` Consensus Attribution quantify reproducible consensus risk associations rather than causal treatment effects.

7. Independent 2019 Temporal Holdout Validation

Evaluates out-of-time model transportability (`2016-2018` train → `2019` calendar holdout) under secular prescribing shifts with zero recalibration.

8. Claims CPIC Exposure, Not Germline Genotypes

Administrative claims quantify CPIC-actionable medication exposure burden (`pgx_num_drugs`, `pgx_num_cpic_drugs`) rather than inferred patient genotypes.

Methodological Integrity: Eight safeguards unite clinical governance with epidemiological rigor. Crucially, **FDA FAERS kinetic convergence** (>90% ADEs within 21 days) provides external validation for our `ed_non_opioid` cohort logic, while strict controls eliminate target leakage.

Questions & Discussion

Technical Appendix Defenses (A1–A7) →

COMMITTEE Q&A BACKUP BANK

Supplemental Technical Evidence

Mathematical derivations, feature audit catalogs, pharmacogenomic scoring matrices, regulatory/HIPAA frameworks, and reproducible notebook pipelines supporting defense inquiries.

🌐 Open Interactive Web Presentation (pgx.jerome-dixon.io/9NOV_presentation) ↗

Appendix A1: Formal Feature Attribution (FFA) — Best-Model CNF & AXP Bounds

SPECIFIC AIM 2 · STEPS 6–8 & 10 | MATHEMATICAL VERIFICATION & PYSAT AXP

Step 7 Dual-Model Gate & Step 8 CNF Compilation

- Step 7 Dual-Model `TreeSHAP` Gate:

$$\mathcal{F}_{\text{SHAP}} = \text{supp}(\text{SHAP}_{\text{CatBoost}}) \cap \text{supp}(\text{SHAP}_{\text{BestXGB}})$$

  Requires cross-architecture agreement before symbolic extraction (since CatBoost CTR hashing precludes direct symbolic path parsing).
- Step 8 CNF Compilation on `best_xgb_variant`:
  - Parses JSON tree dumps of `best_xgb_variant` (`xgb` vs. `xgb_rf` winner by **PR-AUC** then **Recall**).
  - Explodes root-to-leaf paths into tabular intervals (`_explode_tree_to_dataframe`) and compiles Conjunctive Normal Form (CNF) clauses (`rule_clauses`).

SHAP-Guided Minimal Hitting-Set AXP & Complexity

- SHAP-Guided Rule Filter → PySAT `Hitman`:
  - Rather than brute-forcing $> 10,000$ matched tree rules per patient, prioritizes top SHAP-supported matched rules (with stratified rare-population coverage safeguards) into the PySAT `Hitman` minimal hitting-set solver to extract minimal **Abductive Explanations (AXPs / Horn clauses)** with $\text{Supp}(R) \geq 0.05$ and $\text{Conf}(R) \geq 0.70$.
- Step 8 Attribution Responsibility & Step 10 Combined Synthesis:

$$\text{AttrResp}(f) = \frac{\text{RuleFreq}(f)}{\sum_k \text{RuleFreq}(f_k)} \times \text{SHAP}_{\text{consensus}}(f)$$

$$\text{Score}_{\text{combined}}(f) = 0.5 \cdot \text{SHAP}_{\text{norm}}(f) + 0.5 \cdot \text{FFA}_{\text{norm}}(f)$$

- **Complexity vs. Neural Nets:** Tree CNF extraction is polynomial $\mathcal{O}(T \cdot L)$, whereas ReLU neural network minimal explanation is **co-NP-complete** (Katz et al., 2017).

↩ Return to Consensus Attribution (Aim 2)

Operational Threshold Validation Protocol (`SHAP ⩾ P75` ∩ `Supp ⩾ 0.05, Conf ⩾ 0.70`): Framed as a tunable *precision–volume dial* for clinical review capacity and validated via a 2-part sensitivity protocol: (1) **Two-Gate Mirror Ablation** (`SHAP-only` vs. `FFA-only` vs. `SHAP ∩ FFA` grid sweep testing top-K core signal retention, boundary noise filtering, and review-load compression); and (2) **Utilization-Free Specification Stress Test** (stripping utilization covariates/density strata to compare univariate `SHAP` Jaccard rank stability vs. symbolic `FFA` rule persistence under model perturbation).

Appendix A2: Step 3b Target Leakage — Temporal Event-Partitioning Protocol

SPECIFIC AIM 1 · STEP 3B | TEMPORAL PROCESS AUDIT

Leakage Vector Category	Representative Claims Signals	Temporal Timing	Clinical Mechanism
Post-Event OUD Rescue & MAT Induction	Buprenorphine, Suboxone, Naloxone, Methadone, Extended-Release MAT	$t \geq t_0$ (Post-Index)	Initiated in ED or bridge clinic *as a consequence following* acute overdose
Confirmatory Toxicology & Monitoring	Presumptive & definitive urine drug screens (e.g., CPT 80307, 80305, G0480)	$t \geq t_0$ (Post-Index)	Ordered during acute ED triage or post-discharge MAT compliance monitoring
Secondary Administrative Follow-Up	Long-term drug therapy status, rehabilitation & subsequent encounter billing	$t \geq t_0$ (Post-Index)	Downstream administrative claims generated directly by the index visit

Step 3b Formal Event-Partitioning Rule & Symmetric Whitelist

- **Post-Target Ratio Filter:** $\tau_{\text{post}}(f) = \frac{N(t_{\text{claim}} \geq t_0)}{N(t_{\text{claim}} < t_0) + N(t_{\text{claim}} \geq t_0)} \geq 0.80 \implies \text{Purge from } \texttt{safe_feature_filter.json}$.
- **Symmetric Pre-Index Enforcement:** All retained features satisfy $t_{\text{claim}} < t_0$ on $[t_0 - 365, t_0 - 1]$ (Opioid) or $[t_0 - 21, t_0 - 1]$ (Polypharmacy) across cases and 1:5 matched controls.

Why Unmonitored Gain Rankings Fail (`Step 3a` vs. `Step 3b`)

- **The Literature Failure (30% of PROBAST Models):**
  - Unfiltered feature selection loads top tree splits onto post-overdose MAT induction and ED toxicology screens, inflating apparent discrimination by 5–20 AUROC points.
- **Candidacy vs. Defense Scope Demarcation:**
  - Today (`9 NOV`): Lock and verify the **Step 3b temporal event-partitioning methodology** and antecedent whitelist.
  - Final Defense (`23 NOV`): Present complete empirical feature audit counts and 2019 holdout discrimination deltas.

↩ Return to Target Leakage Defense

Appendix A3: Step 5 CPIC Drug Mapping & Patient-Level Count Features

SPECIFIC AIM 1 · STEP 5 (`5_pgx_analysis`) CLAIMS EXPOSURE VS. GENOTYPE DEFENSE

1. Global APCD Drug → CPIC Guideline Mapping (`build_global_drug_cpic_mapping.py`)

- **Official CPIC Reference Corpus:** Ingests `cpic_gene-drug_pairs.xlsx` / `cpic_drug_list.json` (`573` gene–drug pairs, `300` unique drugs, `121` pharmacogenes across CPIC Levels A–D).
- **Automated Drug Vocabulary Matching:** Extracts unique APCD drug features (`DRUG:` , `item_drug_*`) from Step 3 feature importance outputs and matches them against CPIC drug names ($\geq 95\%$ fuzzy threshold) to build `drug_cpic_mapping_global.csv` (`s3://pgxdatalake/gold/pgx_features/global/`).
- **Binary Guideline Indicator:** Determines whether or not each dispensed APCD generic medication has an official CPIC pharmacogenomic guideline ($d \in \mathcal{C}_{\text{CPIC}}$).

2. DuckDB Patient-Level Drug & CPIC Counts (`create_pgx_features_patient_level.py`)

- **Per-Patient Count Features (`mi_person_key`):** `pgx_num_drugs` $_i = |D_i|$ (total distinct pre-index drugs) and `pgx_num_cpic_drugs` $_i = |\{d \in D_i : d \in \mathcal{C}_{\text{CPIC}}\}|$ (distinct CPIC-guideline drugs).
- **Cohort Exposure Counts:** Aggregates distinct pre-index drugs (D_i) per patient in DuckDB and records cohort-level counts of patients with any drug (`pgx_num_drugs` $_i > 0$) vs. patients carrying CPIC-guideline drugs (`pgx_num_cpic_drugs` $_i > 0$).

3. Claims-Derived Exposure vs. Genotype Imputation Defense

- **Step 5 Supervised Model Pipeline (`5_pgx_analysis`):**
  - Uses **observable CPIC drug counts only** (`pgx_num_drugs` , `pgx_num_cpic_drugs`) merged via `add_pgx_features_to_model_data.py` .
  - **Claims Boundary:** Administrative claims quantify dispensed drug exposure, not molecular genotypes or germline DNA sequences.
- **Strict Exclusion of Race-as-Proxy Imputation:**
  - Administrative race codes are strictly excluded as an unscientific, inaccurate proxy for genetic ancestry or metabolizer status.
  - Preserves methodological integrity: models test whether actionable medication burden predicts ED risk without asserting unmeasured biology.
- **Scope Boundary (Candidacy vs. Final Defense):**
  - Candidate models evaluate observable claims exposure directly; interactive downstream point-of-care patient genotype and diplotype cards are staged for the **November 23 Final Dissertation Defense**.

↩ Return to Claims Join & Schema ↩ Return to Model Selection

Appendix A4: Computational Pipeline & Notebook Architecture

AIM 1 AIM 2 AIM 3 FULL TRACEABILITY

Jupyter Notebook Pipeline Directory (`pgx-analysis`)

Notebook	Steps	Core Functions & Deliverables
`0_config_and_pipeline.ipynb`	Setup	Environment config, S3-to-NVMe sync, DuckDB data dictionary
`1_cohort_workflow.ipynb`	1a, 1b, 2	FAERS & APCD 21-day ADE window calibration, dual cohort extraction, 1:5 incidence-density matching, 0-day guard
`2_feature_importance.ipynb`	3a, 3b, 3c	Feature screening, Step 3b BupaR event-partitioning audit, safe antecedent whitelist
`3_model_train_shap_ffa.ipynb`	4, 5, 6, 7, 8	Density tiers, CPIC drug mapping & counts (`pgx_num_cpic_drugs`), Step 6 best-model selection, TreeSHAP & FFA rules
`3_model_sensitivity.ipynb`	CH4 Supp	Utilization-free sensitivity & claims-density stratification checks
`4_dashboard_visuals.ipynb`	9	BupaR clinical pathways, DTW trajectory clusters, FP-Growth itemsets
`5_build_and_deploy.ipynb`	10	Multi-stage Docker packaging, AWS Lambda deployment, API Gateway

Execution Model, Checkpoints & Reproducibility

- **S3 Checkpoint Idempotency:**
  - Each stage syncs inputs from `s3://pgx-analysis-data/` to local NVMe.
  - SHA-256 state markers prevent redundant compute on completed steps.
- **Dual Compute Environments:**
  - **Local Workstation:** Windows 11, DuckDB 1.4+, Python 3.11, R 4.3 (bupaR).
  - **Batch Compute:** Amazon Linux 2023 EC2 (`mushin` account `535362115856`).
- **Source Control & Public Artifacts:**
  - Monorepo: `github.com/Jerome3590/pgx-analysis`

↩ Return to Dissertation Portfolio

Appendix A5: Specific Aims, Methodological Estimators & Evaluation Metrics

SPECIFIC AIM 1 | SPECIFIC AIM 2 | SPECIFIC AIM 3 | PRE-RESULTS METHODOLOGICAL & MATHEMATICAL SPECIFICATION

Specific Aim & Pipeline Stage	Pre-Specified Design & Transformation	Mathematical Estimator & Governance Gate
Aim 1 · Steps 1–2: Matching & `0-Day` Guard	`1:5` incidence-density control sampling on `(age_band, sex, year)`; lookback $[t_0 - W,\ t_0 - 1]$	$t_{\text{claim}} < t_0 \quad (W = 365\text{d}$ `opioid_ed`$, W = 21\text{d}$ `ed_non_opioid`$)$ (purges $t = t_0$ discharge fills)
Aim 1 · Step 3b: Leakage Ratio $\tau_{\text{post}}(c)$	Event-level `bupaR` pre/post index (t_0) timing partition across candidate claims	$\tau_{\text{post}}(c) = \frac{N_{t \geq t_0}(c)}{N_{t<t_0}(c) + N_{t \geq t_0}(c)} < 0.80$ (locks symmetric `safe_feature_filter.json`)
Aim 1 · Step 4: Density Strata & Burstiness B	Training (`2016-2018`) pre-index claim quantiles ($P_{25},\ P_{50},\ P_{95}$); inter-claim gaps Δt	$b(n_{\text{ev}}) \in \{0,\ 1,\ 2,\ 3\}$ (`low` - `extreme`); $B = \frac{\sigma_{\Delta t} - \mu_{\Delta t}}{\sigma_{\Delta t} + \mu_{\Delta t}} \in [-1,\ 1]$
Aim 1 · Step 5: CPIC Drug Mapping & Counts	Global APCD drug → CPIC guideline mapping (`drug_cpic_mapping_global.csv`) & patient CPIC drug counts	`pgx_num_drugs`$_k = \lvert D_k \rvert$; `pgx_num_cpic_drugs`$_k = \lvert \{d \in D_k : d \in \mathcal{C}_{\text{CPIC}}\} \rvert$
Aim 1 · Step 6: MCCV Ensemble & Holdout	25-split stratified MCCV (`CatBoost`, `XGB`, `XGB-RF`) + Platt scaling on untouched `2019` holdout	$\hat{p}_{\text{ens}}(\mathbf{x}) = \sum_{m=1}^{3} w_m \hat{p}_m(\mathbf{x}), \quad w_m = \frac{\text{PR-AUC}_m}{\sum_j \text{PR-AUC}_j}$; $\text{Lift} = \frac{\text{AUPRC}}{\pi}, \quad \text{BS} = \frac{1}{N}\sum (\hat{p}_k - y_k)^2, \quad \text{ICI}$

Specific Aim & Attribution Stage	Pre-Specified Design & Demarcation	Mathematical Estimator & Governance Gate
Aim 2 · Steps 7–8: Consensus Gate $\mathcal{C}$ ★	Dual-model `TreeSHAP` (`Cat ∩ Best XGB`) + PySAT `Hitman(m22)` minimal hitting-set `AXP`	$\overline{\lvert \phi_j \rvert} \geq Q_{75} \ \cap\ \text{Supp}_{\text{AXP}}(f_j) \geq 0.05$ $\cap\ \text{Conf}(r) = \frac{N(r \wedge \hat{y}=1)}{N(r)} \geq 0.70$
Aim 2 · Step 8: Intervention Rate (`IR`) ★	In-model consensus risk sensitivity contrast ($f_i \to x_{\text{med}}$, non-causal); **never from FP-Growth**	$\text{IR}_{\Delta\hat{p}}(f_i) = \frac{1}{n}\sum_{k=1}^{n}\left(\hat{p}^{(k)} - \hat{p}^{(k)}_{f_i \to \text{med}}\right)$; $\text{IR}_{\text{AXP}}(f_i) = \frac{1}{\lvert E_1 \rvert}\sum_{k \in E_1} \mathbb{I}\left(\hat{y}_{f_i \to \text{med}} = 0\right)$
Aim 2 · Step 8: Interaction Effect (`IE`) ★	In-model consensus risk contrast Δ_{11} & Class-1 `AXP` co-occurrence lift (non-causal; 1,000 bootstrap CIs)	$\Delta_{11} = \mathbb{E}[\hat{p}_{11}] - \mathbb{E}[\hat{p}_{10}] - \mathbb{E}[\hat{p}_{01}] + \mathbb{E}[\hat{p}_{00}]$; $\text{IE}(f_i, f_j) = \frac{P(f_i \wedge f_j \mid \text{AXP}_1)}{P(f_i \mid \text{AXP}_1)\,P(f_j \mid \text{AXP}_1)} > 1.0$
Aim 3 · Step 9: Visual Analytics Demarcation	Training-cohort `bupaR`, `DTW`, & `FP-Growth`; **strictly excluded from predictor matrix**	$D_{\text{DTW}} = \min_{\pi \in \mathcal{A}_w} \sqrt{\sum \lVert \mathbf{x}_{i,t_1} - \mathbf{x}_{j,t_2} \rVert_2^2}$; $\text{Lift}_{\text{FPG}}(A \to B) = \frac{\text{Supp}(A \cup B)}{\text{Supp}(A)\,\text{Supp}(B)}$
Aim 3 · Step 10: OODA Scenario Delta $\Delta\hat{p}$	Stateless consensus risk scenario delta ($\mathbf{x} \to \mathbf{x}'$); Imputation of Normality ($x_{\text{unobs}} = 0$)	$\Delta\hat{p} = \hat{p}_{\text{ens}}(\mathbf{x}') - \hat{p}_{\text{ens}}(\mathbf{x})$; $\text{Score}_{\text{comb}}(f) = 0.5 \cdot \text{SHAP}_{\text{norm}}(f) + 0.5 \cdot \text{FFA}_{\text{norm}}(f)$

Candidacy (`9 NOV`) vs. Final Defense (`23 NOV`) Scope Demarcation: Today's examination locks the pre-specified study designs, mathematical estimators, and strict predictive-vs.-visual boundary across Specific Aims 1–3. Complete empirical 2019 holdout outcomes, feature-level `IR` / `IE` rankings, and post-execution sensitivity findings are staged for the November 23 Final Dissertation Defense.

↩ Return to Consensus Attribution (Aim 2) | ↩ Return to Methodological Fences | Appendix A6: Master Acronym Reference →

Appendix A6: Master Acronym & Technical Nomenclature Reference

GOVERNANCE: CRISP-DM · TRIPOD+AI · OODA | APCD · HCG (MILLIMAN) · CPIC PGX | GBDT · TREESHAP · PYSAT FFA/AXP | SERVERLESS CDS & MTM

Acronym	Full Expansion	Operational Role in Dissertation
`CRISP-DM`	Cross-Industry Standard Process for Data Mining	6-phase systems engineering lifecycle governing Steps 1a–10 (Business Understanding → Data Prep → GBDT Modeling → Lambda Deployment).
`TRIPOD+AI`	Transparent Reporting of a Multivariable Prediction Model (+ AI)	27-item biomedical validation standard enforcing Type 2b/3 temporal split (`2016-2018` train → untouched `2019` holdout; zero recalibration).
`OODA` **Loop**	Observe–Orient–Decide–Act Loop	Clinical workflow boundary: GBDT & Horn rules *Observe/Orient*; human pharmacists retain 100% authority to *Decide/Act*.
`PROBAST` / `SQLR`	Prediction Model Risk of Bias Assessment Tool / Systematic Quant. Lit. Review	Chapter 1 PRISMA 2020 review ($n = 94$ ML studies; $n = 73$ PROBAST models) exposing the **"Empty Intersection"** ($n = 0/73$).
`APCD` / `VHI`	All-Payer Claims Database / Virginia Health Information	Statewide longitudinal medical & adjudicated pharmacy claims lake (`2016-2019` , 4.2M patients, 380M claims, deterministic `mi_person_key`).
`HCG` **(Milliman)**	Milliman Health Cost Guidelines (HCG Grouper)	Standardized APCD encounter acuity taxonomy (`hcg_setting` , `hcg_line` , `hcg_detail`) anchoring ED visits: `P51` / `P51b` (PHY ED), `O11` (ER), `P33` (Urgent Care).
`CPIC` / `PGx`	Clinical Pharmacogenetics Implementation Consortium / Pharmacogenomics	Step 5 maps APCD drugs to CPIC guideline status (`drug_cpic_mapping_global.csv`) to quantify patient-level CPIC medication exposure burden (`pgx_num_drugs` , `pgx_num_cpic_drugs`).
`ADE` / `DDI` / `FAERS`	Adverse Drug Event / Drug–Drug Interaction / FDA Adverse Event Reporting System	Geriatric **"Silent ADEs"** (`R29.6` falls, `S72` hip fx, `R55` syncope, `N17` AKI) & FDA FAERS 21-day onset kinetics ($> 90\%$ acute ADEs).
`OUD` / `MME` / `MAT`	Opioid Use Disorder / Morphine Milligram Equivalents / Meds for Addiction Tx	12-month chronic opioid trajectory anchors (`F11.20` , `T40.0X-T40.4X`) vs. post-index rescue drugs purged by Step 3b (buprenorphine, naloxone).

Acronym	Full Expansion	Operational Role in Dissertation
`GBDT` / `XGB-RF`	Gradient-Boosted Decision Trees / XGBoost Random Forest	Piecewise-constant ensembles comparing oblivious trees (`CatBoost`), depth-wise trees (`XGBoost`), and bagged trees (`XGBoost-RF`).
`MCCV` / `TPE`	Monte Carlo Cross-Validation / Tree-structured Parzen Estimator	25-split stratified development CV & Bayesian hyperparameter optimization (`Optuna` , 50 trials) with Platt probability calibration.
`AUROC` / `PR-AUC` / `ICI`	Area Under ROC / Precision-Recall AUC / Integrated Calibration Index	Discrimination, imbalanced minority-class precision-recall lift, Brier Score, ICI calibration, and Decision Curve Analysis (`DCA`).
`XAI` / `TreeSHAP`	Explainable AI / Tree SHapley Additive exPlanations	Exact game-theoretic feature attribution (ϕ_j) intersected across `CatBoost` ∩ `Best XGBoost` ($\overline{\lvert\phi_j\rvert} \geq Q_{75}$) in Step 7.
`FFA` / `CNF` / `PySAT`	Formal Feature Attribution / Conjunctive Normal Form / Python SAT Solver	Step 8 compilation of tree paths into CNF clauses solved via PySAT `Hitman` minimal hitting sets in polynomial $\mathcal{O}(T \cdot L)$ time.
`AXP` **& Horn Rules**	Abductive Explanation & Boolean Horn Logic Clauses	Minimal irreducible `IF-THEN` conditions ($P_1 \wedge P_2 \implies \hat{y} = 1$) guaranteeing Class-1 risk ($\text{Supp} \geq 0.05, \text{Conf} \geq 0.70$).
`IR` **&** `IE` **(**Δ_{11}**)**	Intervention Rate & Interaction Effect (2nd-Order Risk Contrast)	In-model consensus risk attenuation drop ($\text{IR}_{\Delta\hat{p}}, \text{IR}_{\text{AXP}}$) and cross-partial synergy lift ($\Delta_{11}, \text{IE} > 1.0$) strictly from trees/AXPs.
`bupaR` / `DTW` / `FPG`	Business Process Analysis in R / Dynamic Time Warping / FP-Growth	Exploratory process flows, trajectory distance (D_{DTW}), and itemset lift (Lift_{FPG}) **strictly excluded from supervised GBDT matrices**.
`CDS` / `MTM` / `PHI`	Clinical Decision Support / Medication Therapy Mgmt / Protected Health Info	Aim 3 stateless, zero-PHI AWS Lambda microservice ($\mu_{\text{warm}} = 6\text{ms}, < \$5/\text{mo}$) driving an asynchronous Tier 1–3 pharmacist review queue.

↩ Return to Governance and Frameworks | ↩ Return to Polypharmacy Cohort Pipeline | ↩ Return to Methodological Fences | Appendix A7: Regulatory & HIPAA Scope →

Appendix A7: Regulatory Scope Demarcation & HIPAA Cloud Architecture

SPECIFIC AIM 3 · CLINICAL DEPLOYMENT & TRANSLATION | REGULATORY SCOPE: FTC ACT VS. HIPAA BAA | AWS HIPAA-ELIGIBLE ENCLAVE & SHARED RESPONSIBILITY

1. Scope Demarcation: Direct Patient vs. Hospital Deployment

- **Direct Patient Self-Use (FTC Consumer Scope):**
  - Direct individual input of personal health or genetic data is governed by **FTC Act Section 5** and the **Health Breach Notification Rule (HBNR)**.
  - HIPAA applies *exclusively* to covered entities and business associates; when a consumer directly tool, a BAA is **neither required nor legally valid**.
- **Hospital / Healthcare Organization Use (HIPAA Scope):**
  - Clinical deployment processing patient records constitutes ePHI under **45 CFR § 160.103**.
  - **Two-Tier Legal BAA Execution Chain:**

$$[\text{ Covered Entity / Hospital }] \xrightarrow{\text{BAA \#1}} [\text{ Our Legal Entity }] \xrightarrow{\text{BAA \#2}} [\text{ AWS via AWS A}$$

  - Entity executes BAA with hospital; entity executes BAA with AWS via AWS Artifact.
- **VCU CCTR Pilot Grant Integration:**
  - VCU Wright Center CCTR $25,000 Grant formally funds Aim 2 Compliance & Legal Review ($4,50 partner with VCU Health Compliance and Counsel to execute institutional DUAs and BAAs prior to prospective clinical pilot testing.

2. AWS Technical Reality: "HIPAA-Eligible" vs. "HIPAA-Compliant"

- **AWS Shared Responsibility Model:**
  - AWS services are *HIPAA-eligible*; application architecture must implement mandatory administrative, physical, and technical safeguards.
- **Mandatory Cloud Technical Configurations:**
  - **S3 Buckets:** Enforce AWS KMS encryption (`SSE-KMS`), block all public access, enforce TLS (`aws:SecureTransport`) in bucket policies.
  - **AWS Lambda & Memory Security:** Environment variables KMS-encrypted; Firecracker microVM `/tmp` wiped post-invocation; zero ePHI persisted.
  - **CloudWatch Log Scrubbing (Critical):** Custom log sanitizers strictly prevent ePHI from leaking into log streams, stdout, or telemetry.
  - **CloudTrail Auditing:** Immutable management and data API event trails under **45 CFR § 164.312(b)**.
- **AI & External API Boundaries:**
  - **AWS-Native:** Amazon Bedrock / SageMaker models hosted within VPC fall under AWS BAA.
  - **External Third-Party APIs (OpenAI, Anthropic):** Inbound/outbound calls require a separate signed BAA with each vendor, regardless of ephemeral processing.

↩ Return to Aim 3 Decision Support (Slide 8) | ↩ Return to Methodological Fences (Slide 10)
Technical Appendix Overview →

Architectural Synthesis: Ephemeral compute with zero persistent PHI is a sound engineering safeguard, but enterprise clinical deployment legally requires the **two-tier BAA chain** and AWS HIPAA hardening. VCU CCTR Pilot Grant funding guarantees institutional legal review and DUA execution before prospective clinical pilot deployment.